

## Calculation and interpretation of p-values

# Presentation Outline

- 1 Calculating the p-value: randomization vs. permutation test
- 2 Interpretation of a p-value
  - Where did 0.05 come from?

# Permutation tests

1) Permutation test: Enumerate all possible random assignments of treatments

- E.g., 5 individuals, 3 get treatment A, 2 get treatment B

| Individual |   |   |   |   |
|------------|---|---|---|---|
| 1          | 2 | 3 | 4 | 5 |
| A          | A | A | B | B |
| A          | A | B | A | B |
| A          | A | B | B | A |
| A          | B | A | A | B |
| A          | B | A | B | A |
| B          | A | A | A | B |
| B          | A | A | B | A |
| A          | B | B | A | A |
| B          | A | B | A | A |
| B          | B | A | A | A |

## Permutation tests - 2

- If the test statistic is a difference of means with an observed value = 5
  - Count  $R = \#$  permutations with a random test statistic  $\geq 5$  or  $\leq -5$
  - and  $N = \#$  permutations
  - $p\text{-value} = R / N$
- Notice that the actual study is one of the possible permutations
  - So the smallest possible value of  $R$  is 1
- And the smallest possible p-value is  $1/N$

# Randomization tests

## 2) Randomization test: Random sample of treatment assignments

- $p = (R + 1) / (N + 1)$
- Why the + 1?
  - Random set of treatment assignments unlikely to include actual one
  - When  $H_0$  true, actual assignment is one more sample from the null distribution
  - Consequence is that p value is never 0
  - Smallest possible p value is  $1 / (\# \text{ randomizations} + 1)$
  - Explains why commonly see 999 or 9999 randomizations

# What does a p-value “mean”?

The p-value answers the question “How unusual are the data given a null hypothesis?”

- Consider a test of  $\mu_A - \mu_B = 0$
- The recommended interpretations are:

|                       | Interpretation                       |
|-----------------------|--------------------------------------|
| $0.10 < p$            | No evidence of a difference          |
| $0.05 < p \leq 0.10$  | Weak evidence of a difference        |
| $0.01 < p \leq 0.05$  | Evidence of a difference             |
| $0.001 < p \leq 0.01$ | Strong evidence of a difference      |
| $p \leq 0.001$        | Very strong evidence of a difference |

- There are others

|               |                                        |
|---------------|----------------------------------------|
| $p > 0.05$    | Difference not significant             |
| $p \leq 0.05$ | A statistically significant difference |

# Where did 0.05 come from?

- History is clear that Fisher never intended there to be a cliff
  - Intended the p-value as one component in interpreting study results
  - Other components: the magnitude of the difference, its practical importance, and the scope of the study
- But Fisher was asked what constitutes a small p-value?
  - Informal discussion, probably over afternoon tea
- Fisher's off-the-cuff answer was something like "oh, 1 in 20", i.e., 5%
- No mathematical justification

## Where did 0.05 come from? - 2

Neyman-Pearson approach to hypothesis tests

- Based on a significant / not-significant dichotomy
  - Different concept of a hypothesis test
  - Explicit alternative hypothesis
- Context: Manufacturing widgets
  - Need to meet some quality specification, e.g. 99.9% or more of the parts within tolerance
  - Inspect boxes of widgets before shipping them from the factory
  - Test a sample of parts, null is  $< 99.9\%$  meet specification, want to reject this



## Where did 0.05 come from? - 3

- What can go wrong?
  - Box is **defective** but estimate is 99.9999% good  
Box gets shipped but shouldn't be (type I error)
  - Box is **good** but estimate is 98% good  
Box gets rejected but shouldn't be (type II error)
  - Want both error rates to be small
- Rejecting any box with estimate  $< 99.9\%$  is not appropriate
  - Consider boxes that are "just good enough", 99.9% good
  - Estimate  $< 99.9\%$  rejects 50% of the boxes (approximately)
- Math  $\Rightarrow$  a sampling scheme with specified properties
  - Choose sample size and criterion (e.g., reject if estimate  $< 99.95\%$ )
  - so that type I error rate is small and
  - type II error rate is small

# Where did 0.05 come from? - 4

My view: Quality control of widgets is not the same as a scientific investigation

- Widgets:
  - Dealing with a very large number of boxes
  - Need to make a dichotomous decision (ship or reject)
  - Error rates are statements about many boxes
  - Not whether a single decision is correct
- Scientific investigation
  - Dealing with one study
  - and want to make a justifiable conclusion(s) for that single study
  - Should care more about “how big” instead of “not zero” / “not sure”
- Huge, nasty, arguments between Fisher and Neyman

## Where did 0.05 come from? - 5

- Neyman-Pearson approach was dominant for many years
  - Still the foundation of industrial quality control
- But science moving / moved back to Fisher's approach - a scale of evidence
- If you see a paper reporting  $p < 0.05$ 
  - They are following Neyman-Pearson
- My view:  $p < 0.05$  is insufficient - I care whether  $p = 0.049$  or  $< 0.0001$
- Modern approach, more in line with Fisher, is to report the p-value
  - Usually to 2 non-zero digits, e.g. 0.15, 0.032, 0.0018,  $< 0.0001$

# Important Points

## Important Points

- p value:  $P[\text{as or more extreme test statistic given } H_0]$
- Given  $E = \#$  as more extreme out of  $N = \#$  randomizations
  - Permutation test:  $p = R/N$
  - Randomization test:  $p = (R+1)/(N+1)$
- Interpretation of a p-value
  - Distinction between the Neyman-Pearson decision-theoretic perspective
  - And Fisher's less formal approach
  - Modern focus on scale of evidence and reporting actual p-value